

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (E&C) (Credit System) Degree Examinations

January 2023

19ECE11 – ADHOC AND SENSOR NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five Full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	With the help of a neat diagram, explain a ZigBee protocol stack.	7	L*2	1	1
b)	Discuss the energy consumption of a data processing unit of a sensor node with the relevant equations.	8	L2	1	1
c)	Discuss the main header types of 6LoWPAN standard.	5	L2	1	1
2. a)	Explain the architecture of Wireless Hart protocol with the help of neat figure.	8	L2	1	1
b)	Explain any two basic modulation schemes used in the sensor nodes.	7	L2	2	1
c)	Explain the breakdown curve of a MicaZ sensor node.	5	L2	2	1
3. a)	Discuss the technologies available for wireless link establishment in Wireless sensor networks.	8	L2	2	1
b)	Define pathloss. Deduce an expression for pathloss. How it can be represented considering the multipath effects?	7	L3	2	1
c)	Write the various topologies of IEEE 802.15.4 standard in Wireless sensor networks.	5	L2	2	1
Unit – II					
4. a)	Explain the concept of collision in CSMA protocol.	8	L2	3	1
b)	Discuss the CSMA mechanism with the help of diagrams.	7	L2	3	1
c)	What is a synchronizer? Discuss its usage in a network layer protocol.	5	L2	3	1
5. a)	Explain various operational features of S-MAC protocol with the neat timing diagram.	10	L2	3	1
b)	With the necessary timing diagram, explain a B-MAC protocol.	10	L2	4	1
6. a)	Explain the physical layer measurement techniques used in the localization procedures.	10	L2	4	1
b)	Discuss any two ranging techniques in the range based localization techniques.	10	L2	4	1
Unit – III					
7. a)	Write the advantages and disadvantages of star, mesh and hybrid topologies.	8	L2	5	1
b)	With a neat block diagram, explain a body sensor network connected to the hospital network through gateway.	7	L2	5	1
c)	Discuss various sections of the communication protocol stack of WPAN.	5	L2	5	1
8. a)	Explain a IEEE 802.15.1 Bluetooth protocol stack architecture with a neat diagram.	10	L2	5	1
b)	Discuss the principle of operation of inertial micro-generators energy scavenging techniques.	10	L2	5	1

BT*: Bloom's Taxonomy, L*: Level; CO*: Course Outcome; PO*: Program Outcome
